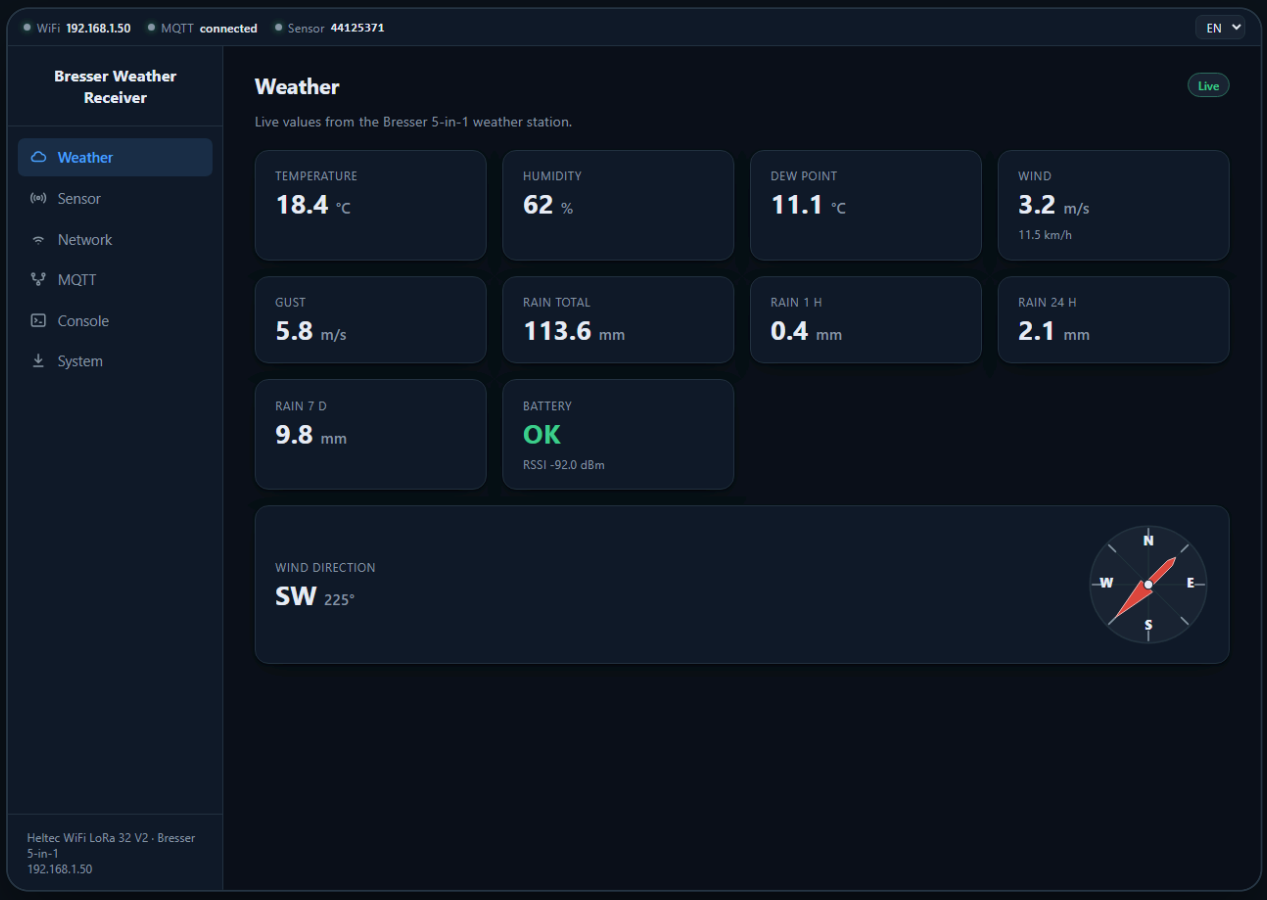
MANUAL

Bresser 5-in-1 Weather Receiver

Receive, display and pass on the measurements of a Bresser 5-in-1 weather station with the Heltec WiFi LoRa 32 V2



Firmware version 1.0.0 · English
github.com/okoebernik/Bresser5in1

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> The screenshots show example data (addresses and names are placeholders) and come from the web interface of firmware 1.0.0.

1. Overview

The receiver is built around a **Heltec WiFi LoRa 32 V2**. Its radio chip (SX1276) receives the telegrams of the Bresser 5-in-1 weather station on 868 MHz. The firmware decodes them, derives further values (dew point, rain of the last hour, 24 hours and 7 days) and makes the data available in three ways:

- on the **OLED display** of the device,
- in a **web interface** on your WiFi (live values, settings, console, firmware update),
- for other devices through **MQTT** or the **REST API** (no MQTT server needed).

The device is a receiver only. It does not transmit anything to the weather station.

2. Requirements

- Heltec WiFi LoRa 32 **V2** (SX1276, 868 MHz variant) with a **868 MHz antenna attached**
- USB cable **with data lines** and a power supply (5 V)
- Bresser 5-in-1 weather station within radio range
- WiFi (2.4 GHz); optionally an MQTT broker
- For the first installation on Windows, if needed the **CP210x** USB driver

3. First installation

For a new or blank board the release contains two files:

File	Purpose
`Bresser5in1-<version>-erstinstallation.bin`	Full flash image (8 MB), written at address `0x0`
`Bresser5in1-<version>.bin`	Firmware only, for later updates in the web interface

Way A: in the browser (no software)

1. Connect the device over USB and open `espressif.github.io/esptool-js` with Chrome or Edge.
2. Choose baud rate `921600`, click *Connect* and select the device's COM port.
3. Enter `0x0` as *Flash Address*, choose the file `…-erstinstallation.bin` and click *Program*.
4. Then restart the device with the **RST** button.

Way B: with esptool

```
esptool --chip esp32 --port COM5 --baud 921600 write_flash 0x0 Bresser5in1-v1.0.0-
erstinstallation.bin
```

For a completely clean start run `esptool --chip esp32 --port COM5 erase_flash` beforehand.

Way C: from source

Arduino IDE with the ESP32 board package (tested: 3.3.10), board *Heltec WiFi LoRa 32(V2)* and the libraries *BresserWeatherSensorReceiver*, *RadioLib*, *WiFiManager*, *PubSubClient*, *ArduinoJson* and *ESP8266 and ESP32 OLED driver for SSD1306 displays*. Open the sketch `Bresser5in1_Merged` and upload it.

> **Note:** The full image erases the configuration storage. WiFi credentials, MQTT settings and the OTA password have to be set up again afterwards. It is built for boards with 8 MB of flash; with only 4 MB please use way C. The detailed guide is in `INSTALLATION.en.md`.

4. First start

1. After flashing, the device restarts. The OLED briefly shows *Bresser Receiver*, then *WS init OK* and the IP address.
2. Without stored WiFi the device opens its own network **BresserHeltec** (no password).
3. Switch your phone or laptop to this network. The setup portal opens automatically, otherwise open `http://192.168.4.1`.
4. Choose *Configure WiFi*, select your own WiFi, enter the password and save.
5. The device connects and shows its **IP address on the OLED** (and in the serial monitor, 115200 baud).
6. Open `http://<IP address>` in the browser, or alternatively `http://BresserHeltec.local`.

The portal stays open for five minutes. If nothing is set up in that time, the device keeps running without WiFi (reception and OLED work). To open the portal again later, hold the **PRG** button (GPIO0) while powering on. The WiFi can also be changed at any time on the *Network* page.

5. The web interface

The interface is reachable at the IP address of the device and adapts to the screen width (on a phone the menu icon opens the sidebar). It consists of a status bar, a sidebar for navigation and the content area.

Status bar and language

The bar at the top shows three indicators: **WiFi** (connected, with IP address), **MQTT** (connected, disconnected or disabled) and **Sensor** (ID of the displayed station). A green dot means fine, yellow a warning, grey inactive. At the top right you switch between **EN** and **DE**; the choice is stored in the browser.

Weather

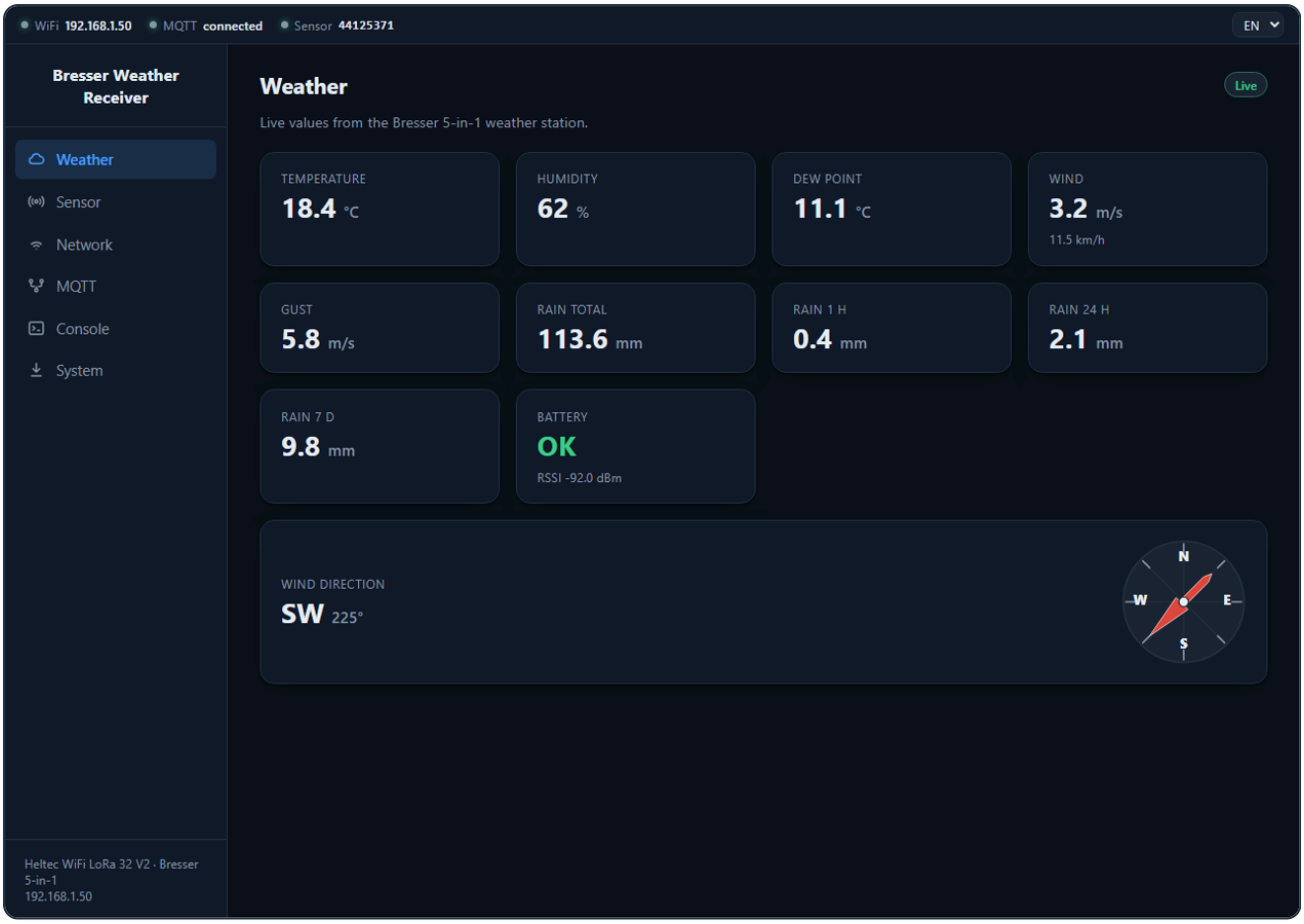


Figure 1: *Weather* page with all live values

The badge at the top right reads *Live* while fresh data arrives, otherwise the age of the last packet or *No data yet*. The page refreshes every three seconds. The station sends new values about every twelve seconds.

Card	Meaning
Temperature, Humidity	Values measured by the station
Dew point	Calculated from temperature and humidity
Wind, Gust	Mean wind speed and gust in m/s; the wind in km/h is shown below
Rain total	Continuous counter of the station in mm
Rain 1 h / 24 h / 7 d	Calculated by the device from the counter. Starts at 0 after a restart
Battery, RSSI	Battery state of the station and signal strength of the last packet in dBm
Wind direction	Compass direction, degrees and wind rose

The total rain counter is protected against radio errors: a jump of more than 5 mm per packet (or a decrease) is only accepted once the next packet confirms the same value. If the station's counter is reset, rain keeps being counted from the reset.

Sensor

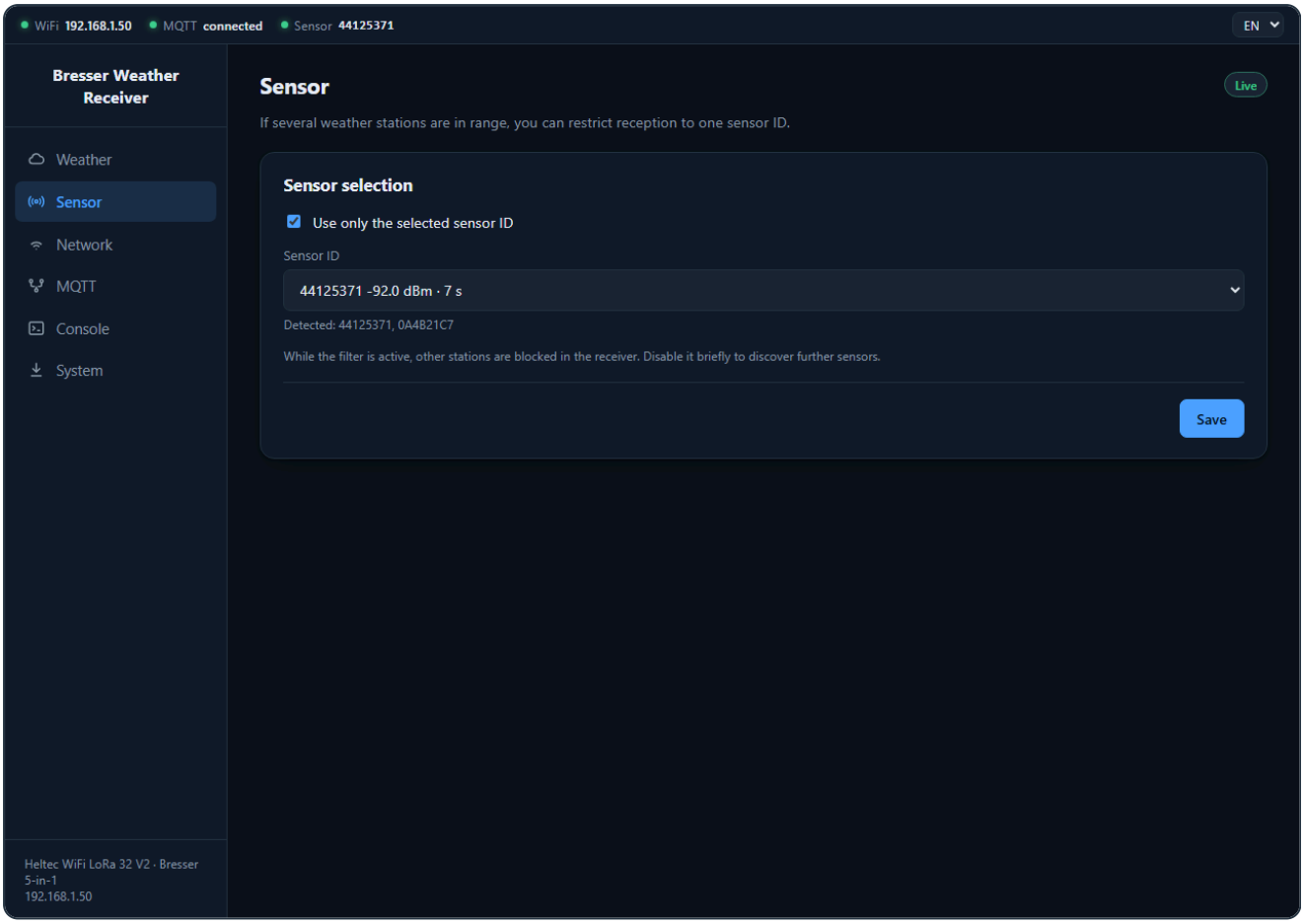


Figure 2: *Sensor* page – choosing the weather station

If several Bresser stations are within range, reception can be restricted to one station:

1. Leave the filter off at first and wait a few minutes until your own station appears in the list.
2. Choose the sensor ID in the selection field. The list shows RSSI and the age of the last packet per station.
3. Switch on *Use only the selected sensor ID* and click *Save*.

While the filter is active, foreign stations are already blocked in the receiver and no longer show up in the list. Switch the filter off briefly to search for further sensors. The dashboard, OLED and rain values always belong to **one** station: the saved sensor ID, otherwise the one received first. Other stations are only published over MQTT (without rain time windows).

Network

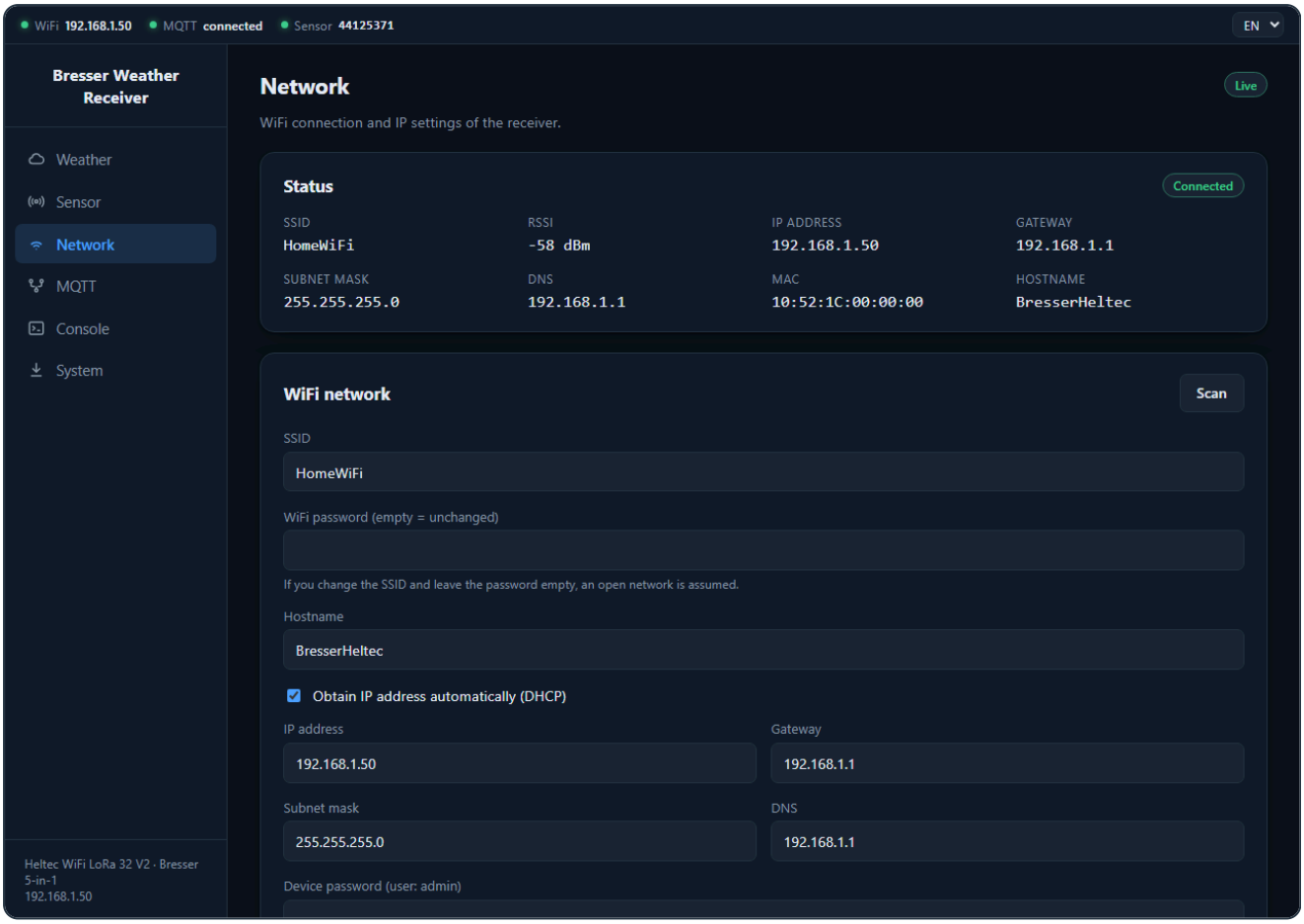


Figure 3: *Network* page – status and settings

The top shows the current state: SSID, WiFi RSSI, IP address, gateway, subnet mask, DNS, MAC address and hostname. Below you can change:

- **WiFi:** *Scan* lists networks in range by signal strength. A click takes over the SSID. Leave the password empty as long as the SSID stays the same; if you change the SSID and leave the password empty, an open network is assumed.
- **Hostname:** letters, digits and hyphen, up to 31 characters.
- **IP address:** automatic via DHCP or fixed (IP, gateway, subnet mask, DNS).

Saving requires the **device password** (user `admin`, password = OTA password). The device restarts afterwards. A new WiFi is tested first and only saved on success; with a wrong password the previous network is kept. After a new network or a static IP the page may no longer be reachable at the old address. The new IP is shown on the OLED. A static IP must fit the network (address, gateway, subnet mask).

MQTT

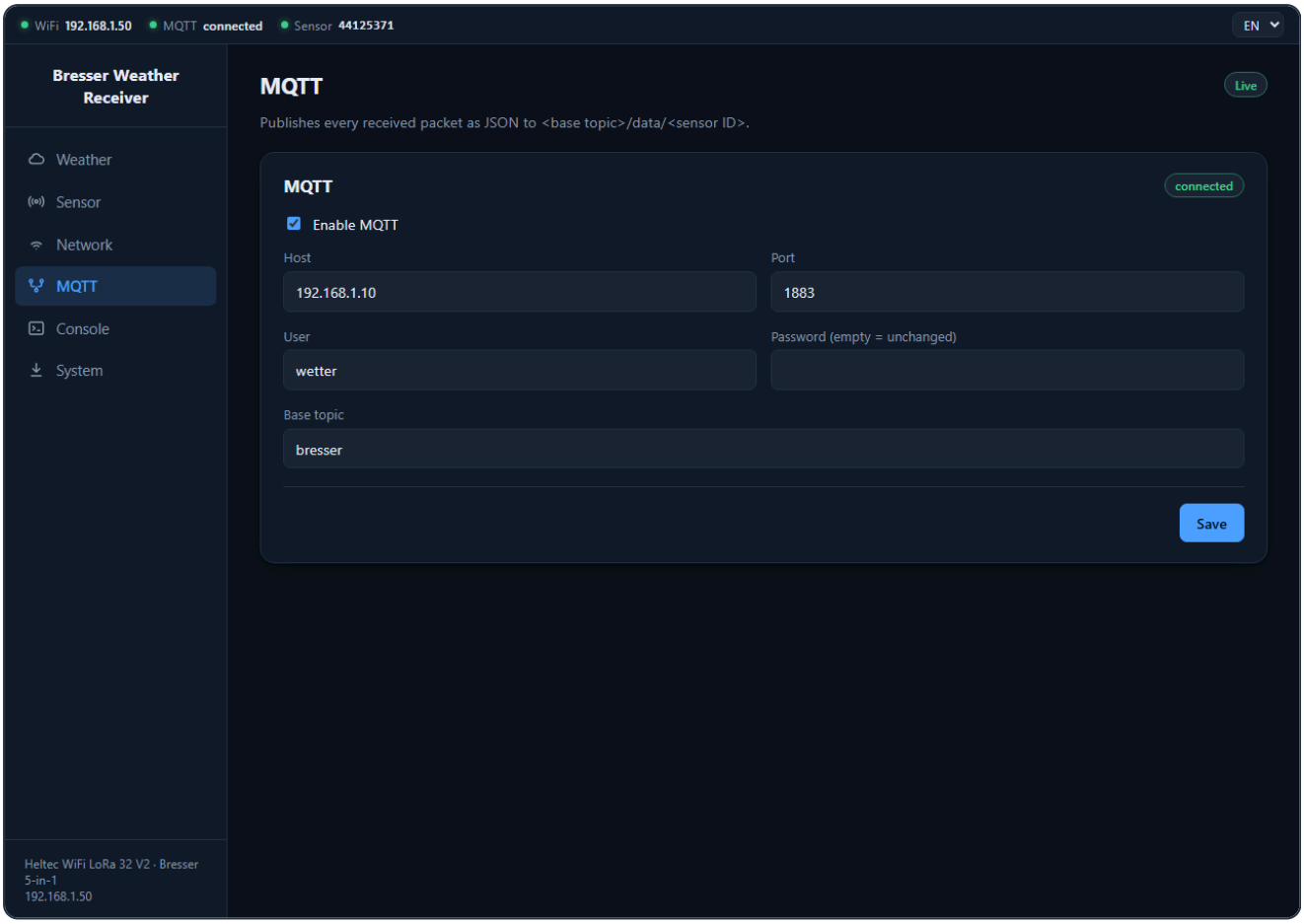


Figure 4: *MQTT* page

MQTT is optional and off by default. Enter host, port (default 1883), user, password and *Base topic* (default `bresser`). The password is never returned to the browser; an empty field leaves it unchanged. The badge shows the connection state. Each station is published under `<base topic>/data/<sensor ID>`, see the chapter "Passing on the measurements".

Console

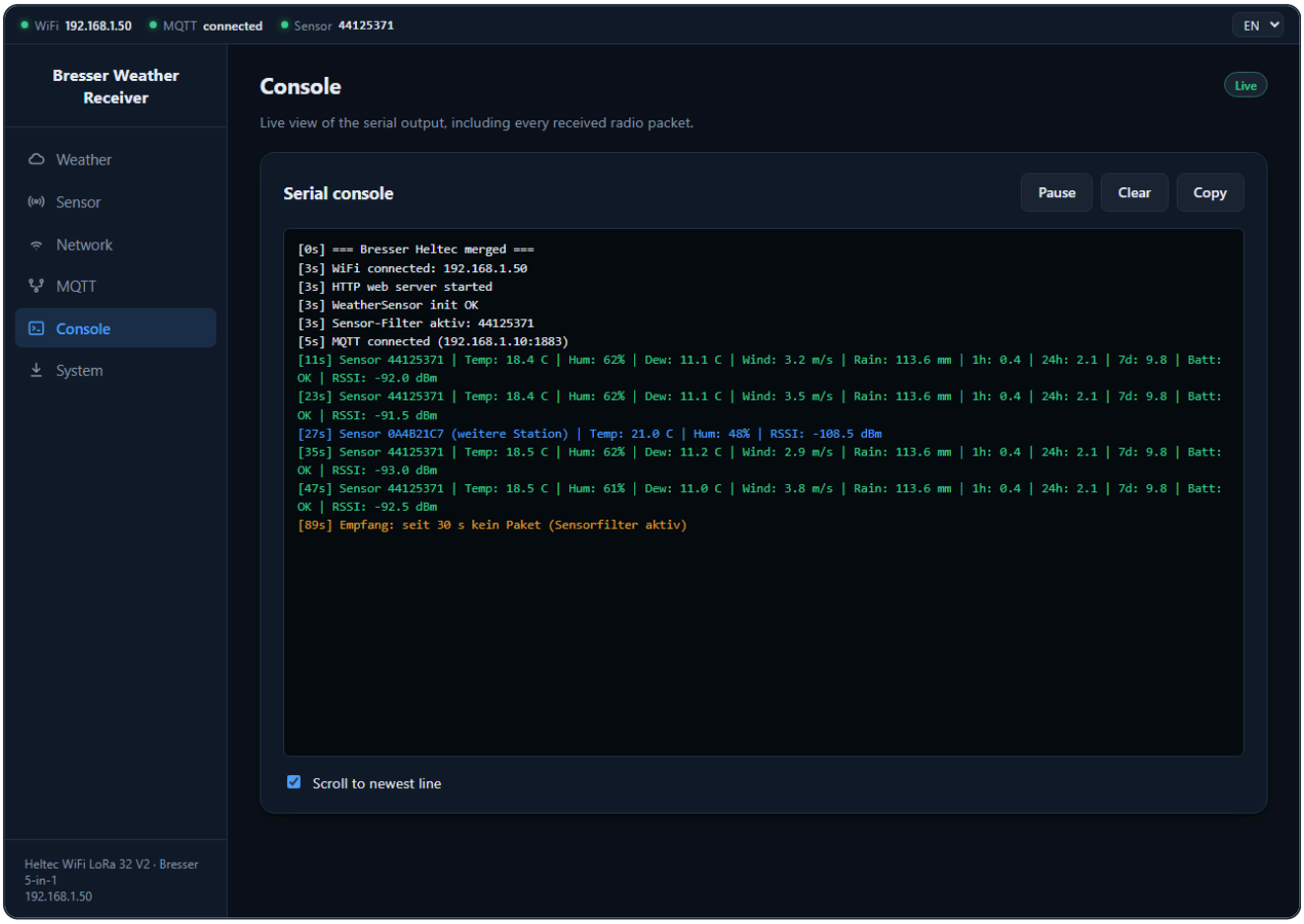


Figure 5: *Console* page with received packets

The console shows the serial output of the device live, without a cable having to be connected. Each line starts with the uptime in seconds. The messages are in German or technical wording and are not translated.

- Green: packet of the displayed station with all values and RSSI
- Blue: packet of another station
- Yellow: warnings, such as "no packet" (every 30 seconds while nothing arrives) or an ignored sensor
- Red: errors

Pause freezes the display, *Clear* empties it, *Copy* puts the text on the clipboard. The device buffers the last 6 KB. The console is the most important tool for judging reception when positioning the device.

System

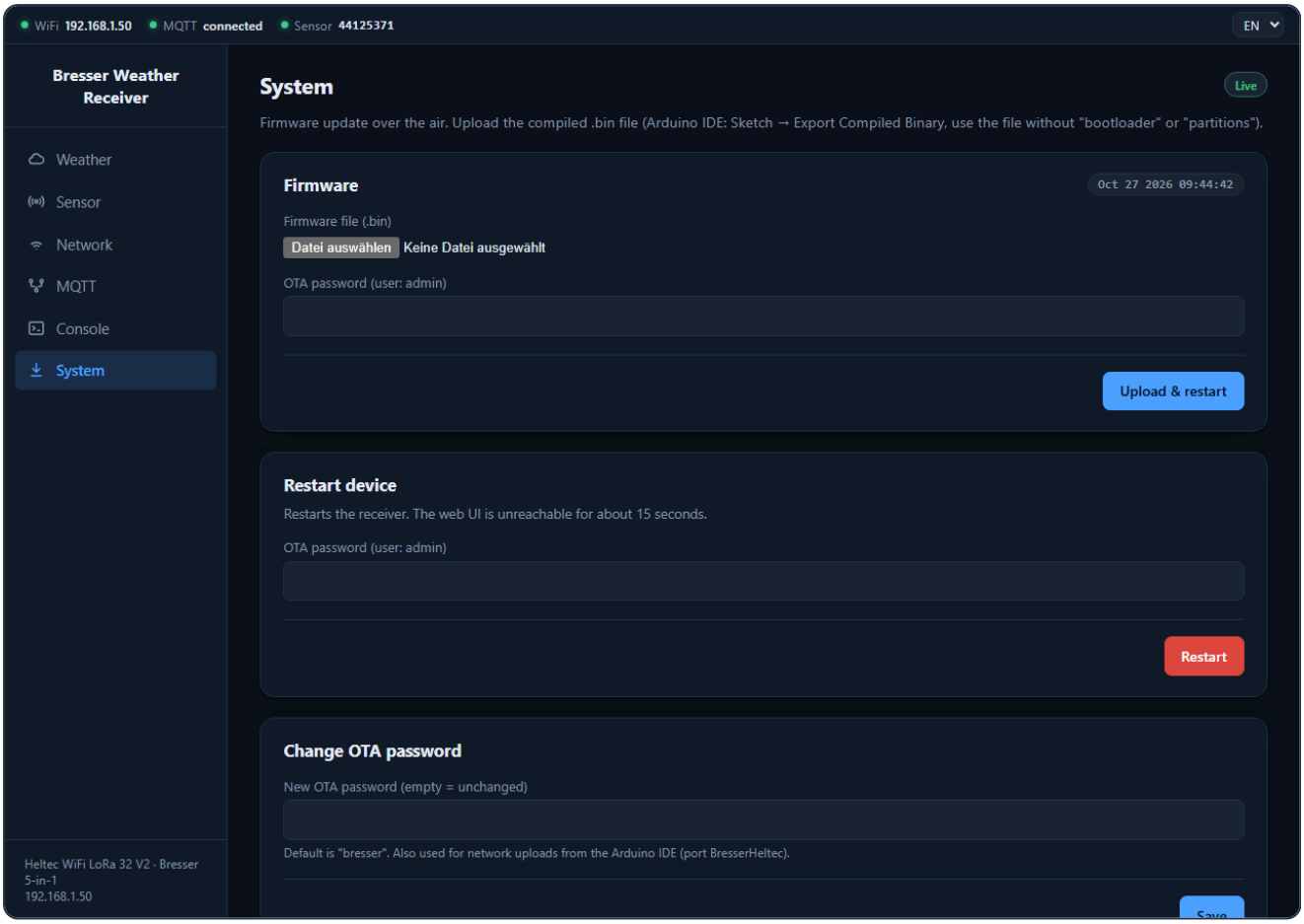


Figure 6: *System* page

Firmware update: choose the file `Bresser5in1-<version>.bin`, enter the OTA password and click *Upload & restart*. A progress bar runs in the browser and on the OLED. Afterwards the device restarts by itself and keeps all settings. The badge at the top shows the build timestamp of the running firmware. If an update is interrupted, the previous firmware stays active. Alternatively the update can be installed from the Arduino IDE (port *BresserHeltec*); a firewall may block this route.

Restart device: restarts the receiver (password required). The interface is unreachable for about 15 seconds afterwards.

Change OTA password: the default password is `bresser` and should be changed right away.

6. OLED display

The display compactly shows, for the displayed station: sensor ID, temperature and humidity, dew point, wind, rain total and rain of the last hour and 24 hours. The image is rotated by 180° compared with the display's default orientation. At startup it shows the IP address; during a firmware update it shows the progress.

7. Passing on the measurements: MQTT and REST API

REST API (no MQTT server)

Other devices, such as a multi-instrument, can query the values directly. The API is read-only, needs no login and sends `Access-Control-Allow-Origin: *`. The pattern matches that of the Clinometer: `GET /api/<device>/values`, flat JSON, unit in the field name, `stale` flag.

```
GET http://<IP>/api/weather/values

{ "valid": true, "stale": false, "sensor_id": "44125371", "age_s": 8,
  "temperature_c": 12.4, "humidity_pct": 95, "dew_point_c": 11.6,
  "wind_speed_ms": 0.0, "wind_gust_ms": 0.0, "wind_dir_deg": 324, "wind_dir_text": "NNW",
  "rain_total_mm": 113.6, "rain_1h_mm": 0.0, "rain_24h_mm": 0.0, "rain_7d_mm": 0.0,
  "battery_ok": true, "rssi_dbm": -106.5 }
```

- `stale` is `true` when no data has been received yet or the last packet is older than 120 seconds.
- Without data the response only contains `valid: false` and `stale: true`; the numeric fields are then missing.
- New values arrive about every 12 seconds. A polling interval of 10 to 15 seconds is enough.

`GET /api/status` returns firmware build, uptime, free memory, WiFi signal, IP, MQTT status, active sensor ID and `last_packet_age_s` (−1 = no packet yet).

MQTT

For every received packet a JSON is published under `<base topic>/data/<sensor ID>`, for example `bresser/data/44125371`:

```
{ "id": 1142051697, "id_hex": "44125371", "temp_c": 12.4, "humidity": 95, "dew_point_c": 11.6,
  "wind_speed_ms": 0.0, "wind_gust_ms": 0.0, "wind_dir_deg": 324, "wind_dir_text": "NNW",
  "rain_mm": 113.6, "rain_hour_mm": 0.0, "rain_day_mm": 0.0, "rain_week_mm": 0.0,
  "battery_ok": true, "rssi": -106.5, "uptime_s": 3600 }
```

Packets of other stations do not contain the fields `rain_hour_mm`, `rain_day_mm` and `rain_week_mm`. To listen in: `mosquitto_sub -h <broker> -t 'bresser/#' -v`.

8. Improving reception

The signal strength (RSSI) is shown on the *Weather* page and in every line of the *Console*. The closer the value is to 0 dBm, the better. In testing, incorrectly decoded packets already appeared at about −105 dBm (for example a constantly changing battery status). That is an experience value, not a manufacturer specification; the better the value, the more reliable the reception.

- **Antenna:** always use a 868 MHz antenna on the IPEX connector and orient it vertically like the station's.

- **Location:** as close as possible to the station, elevated and with a clear line of sight. Walls, metal and window films attenuate strongly.
- **Interference:** keep away from routers, monitors and cheap USB power supplies.
- **Checking:** change the position in small steps and watch the RSSI in the console.

At the new location the device also needs a usable WiFi signal. The *Network* page shows the WiFi signal strength. With weak WiFi the web interface may respond slowly or drop out.

9. Troubleshooting

Problem	Solution
No weather data	Antenna connected? Closer to the station? On the *Sensor* page switch the filter off and check in the *Console* whether any packet arrives.
Wrong station is shown	Switch the filter off, wait, choose the right ID, switch the filter on, save.
Rain 1 h / 24 h / 7 d show 0	Normal after a restart: the windows only fill over time. At startup the first confirmed counter value is the baseline.
Battery alternates between OK and LOW	Usually decoding errors at a weak signal. Improve reception rather than replacing the battery right away.
Web interface not reachable	Read the IP from the OLED or the router, check the network. To set up the WiFi again, hold **PRG** at startup.
Not reachable after a network change	The new IP is shown on the OLED. For a static IP check address, gateway and subnet. In an emergency hold PRG at startup and set up the WiFi again.
MQTT connects but no data arrives	Check host, port, user, password and base topic. Look for "MQTT publish failed" in the *Console*. Check the broker with `mosquitto_sub`.
Update fails	Check the OTA password (error 401 = wrong password). It must be the plain firmware `Bresser5in1-<version>.bin`, not the full image.
Reset everything to factory state	`esptool --chip esp32 --port COM5 erase_flash`, then write the full image again.

10. Security

- The web interface and the API use **unencrypted HTTP**. Operate the device only in a trusted home network, not directly on the internet.
- The **read-only pages and the REST API are reachable without a login**. Anyone on the network can retrieve the weather values. The sensor and MQTT settings can also be changed without a password.

- Network changes, firmware update, restart and changing the OTA password are protected with **HTTP Basic authentication** (user `admin`).
- The default OTA password `bresser` is publicly known. **Please change it right away.**
- The MQTT password is stored in the device but never returned to the browser.

11. Technical data

Property	Value
Hardware	Heltec WiFi LoRa 32 V2 (ESP32, SX1276, 0.96" OLED)
Radio	868 MHz (Europe), receive only
Station	Bresser 5-in-1 (temperature, humidity, wind, gust, wind direction, rain)
Packet interval of the station	about 12 seconds
Web interface	HTTP, port 80, German/English
Firmware update	Over WiFi (two app slots), browser or Arduino IDE
Configuration storage	ESP32 Preferences, survives a restart
Serial interface	115200 baud
Rain windows	1 h, 24 h, 7 d (in RAM, empty after a restart)
Source code & releases	github.com/okoebernik/Bresser5in1